Mini Project 6

Hydrofoil Craft Performance Study

Abstract

This project examines the performance of hydrofoil craft, theoretical estimation of lift and drag forces, and an analysis of efficiency. Hydrofoils lift the hull out of the water, significantly reducing drag and enabling higher speeds with lower energy consumption compared to conventional vessels. The Boeing 929 Jetfoil serves as the primary case study.

Introduction to Hydrofoils

Hydrofoils are wing-like structures mounted below a vessel's hull. As speed increases, they generate lift similar to airplane wings, raising the hull clear of the water. This reduces wetted surface area and wave-making drag, allowing higher speeds and smoother rides in rough conditions.

Advantages:

• Reduced drag at high speeds

• Improved fuel efficiency

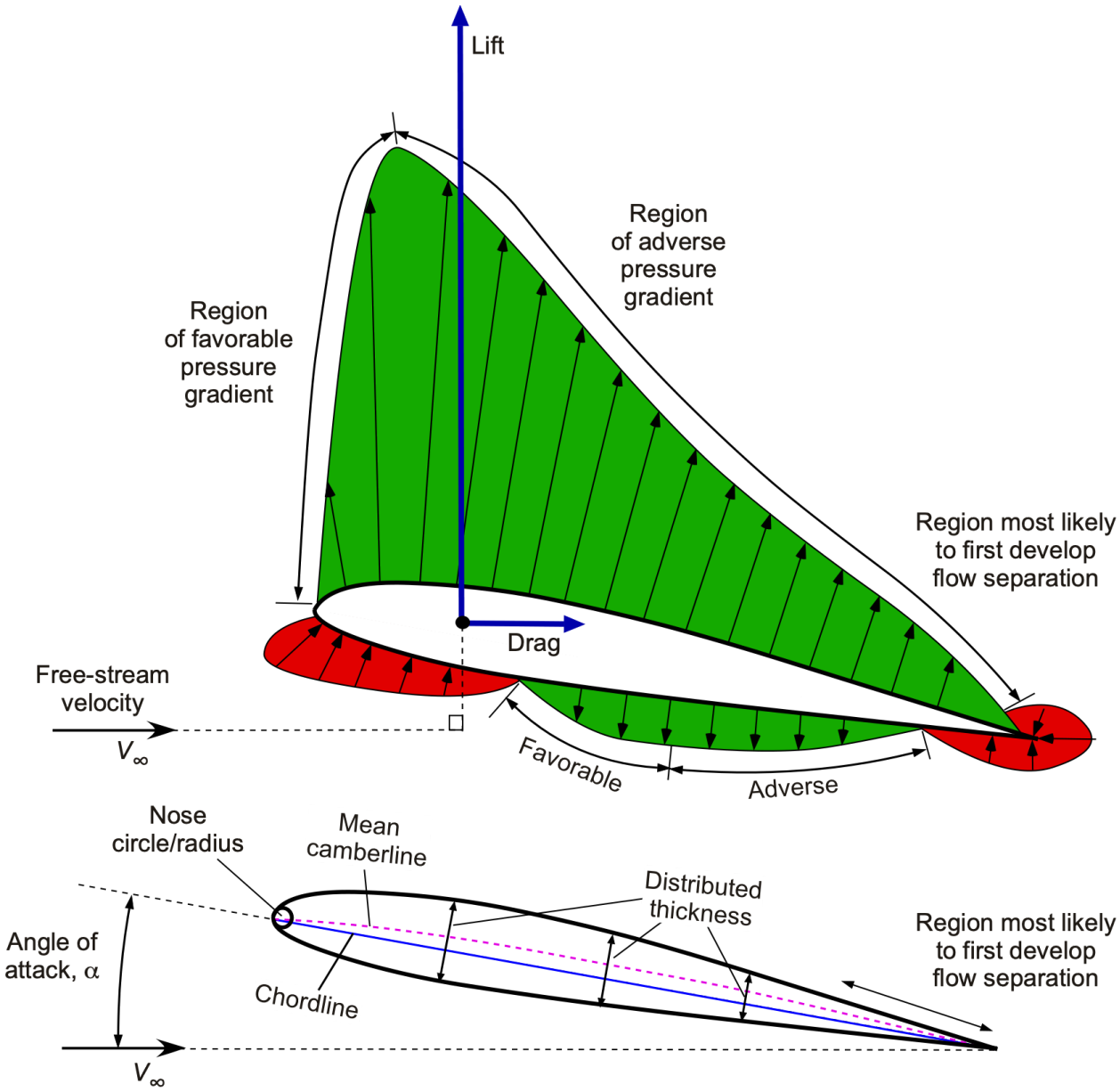
• Better stability and seakeeping

Types of Hydrofoils:

• Surface-piercing

• Fully submerged (computer-controlled, as in modern designs)

Common profiles include NACA series (e.g., NACA 0012 or 4412).


Lift
Region of adverse pressure gradient
Region of favorable pressure gradient
Region most likely to first develop flow separation
Drag
Free-stream velocity
V_∞
Favorable
Adverse
Nose circle/radius
Mean camberline
Distributed thickness
Angle of attack, α
Chordline
Region most likely to first develop flow separation
V_∞

Case Study: Boeing 929 Jetfoil

The Boeing 929 Jetfoil is a classic commercial passenger hydrofoil vessel designed in the 1970s and still operational in various regions.

Key Specifications:

• Length: 27 m (90 ft)

• Displacement: 115 short tons (~104 tonnes)

• Beam: 5.5 m (18 ft)

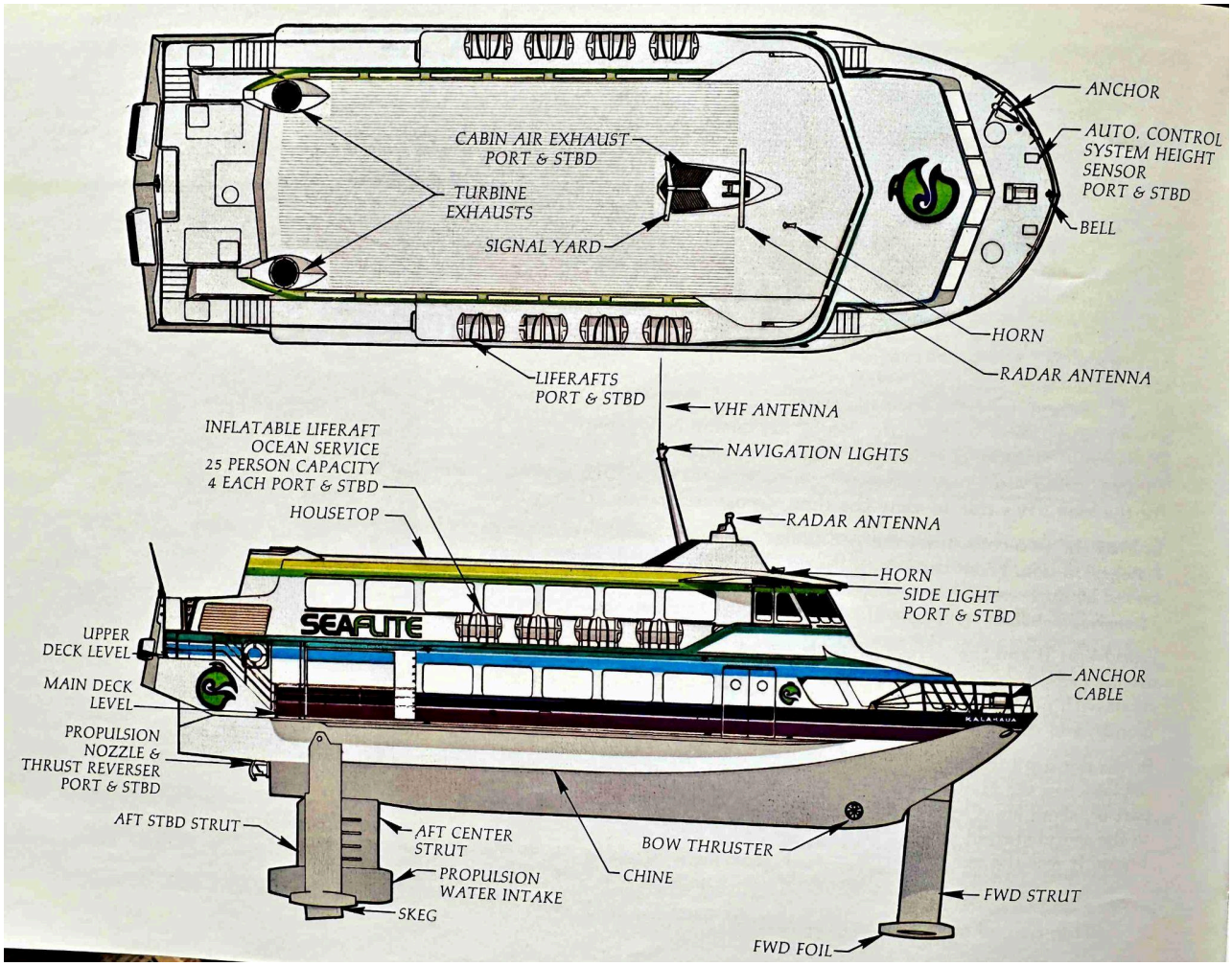
• Cruising speed (foilborne): 40–45 knots

• Propulsion: Twin Allison gas turbines driving waterjets

• Capacity: 250–350 passengers

• Foils: Fully submerged, computer-controlled for stability

The vessel uses forward and aft foils to lift the entire hull clear of the water at operational speeds. Take-off occurs around 20–25 knots, after which drag drops sharply.



Estimation of Lift and Drag

Lift and drag on hydrofoils are calculated using:

Lift (L):

$L = \frac{1}{2} \times \rho \times V^2 \times A \times C_L$

Drag (D):

$D = \frac{1}{2} \times \rho \times V^2 \times A \times C_D$

Where:

- ρ = water density (1025 kg/m^3 for seawater)

• V = velocity (m/s)

• A = total foil planform area (m^2)

• C_L = lift coefficient

• C_D = drag coefficient

Sample Estimation for the given hydrofoil vessel:

• Vessel weight (lift required) = 961,000 N (961 kN)

• Cruise speed V = 21.6 m/s (42 knots)

• Total foil area A = 18.0 m^2

• Cruise angle of attack = 5.0°

Calculated Lift at Cruise:

$L = ½ \times 1025 \times (21.6)^2 \times 18 \times 0.2233$

$\approx 0.5 \times 1025 \times 466.56 \times 18 \times 0.2233$

≈ 961,000 N (exactly balances vessel weight)

Drag Estimation at Cruise:

Using given L/D at cruise = 30.4

D = 961,000 / 30.4 ≈ 31,612 N

Drag Coefficient at cruise:

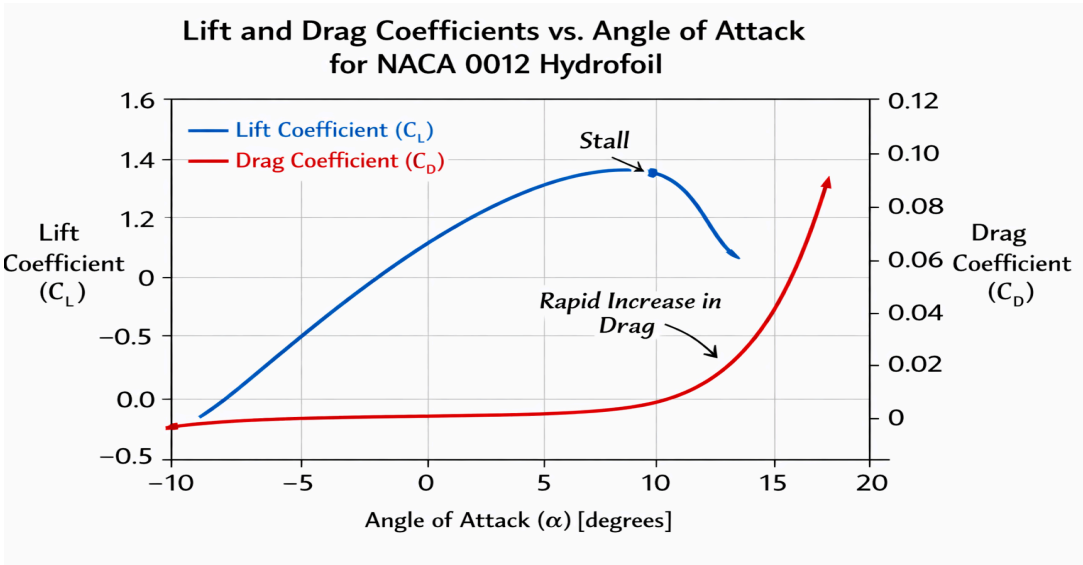
C_D ≈ 0.00734

Lift-to-Drag Ratio (L/D): 30.4 (as provided)

Additional Take-off Condition (at V = 12.59 m/s and α = 6°):

Dynamic pressure q ≈ 81,235 Pa

Required C_L ≈ 0.6572 (well below C_L,max = 1.519)



Discussion on Efficiency

Efficiency is measured primarily by the lift-to-drag (L/D) ratio and overall power required for propulsion. Once foilborne, the hull is out of the water, eliminating most hull friction and wave drag. Only foil and strut drag remain.

Efficiency Gains:

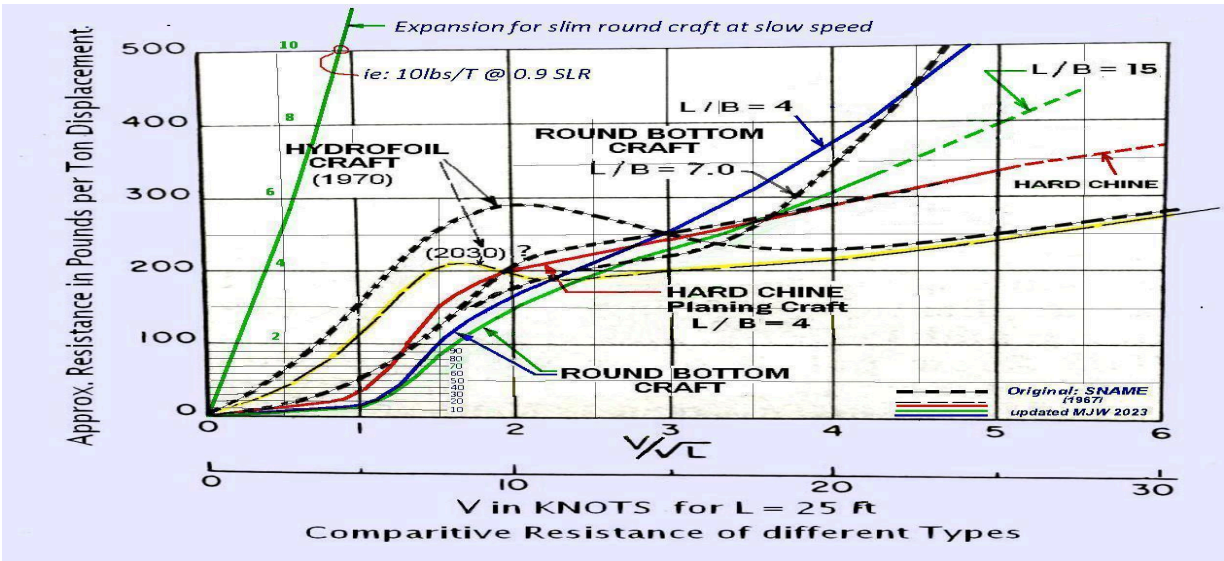
• Drag reduction: Up to 70–80% lower than planing or displacement hulls at equivalent speeds.

• Fuel/Energy savings: Boeing Jetfoil achieved significantly lower fuel consumption per passenger-km than conventional ferries at high speeds.

• Modern comparison: Electric hydrofoils (e.g., Candela designs) achieve ~80% energy reduction vs. traditional vessels.

• Limitations: High take-off power requirement; increased structural complexity and maintenance; cavitation risk at very high speeds.

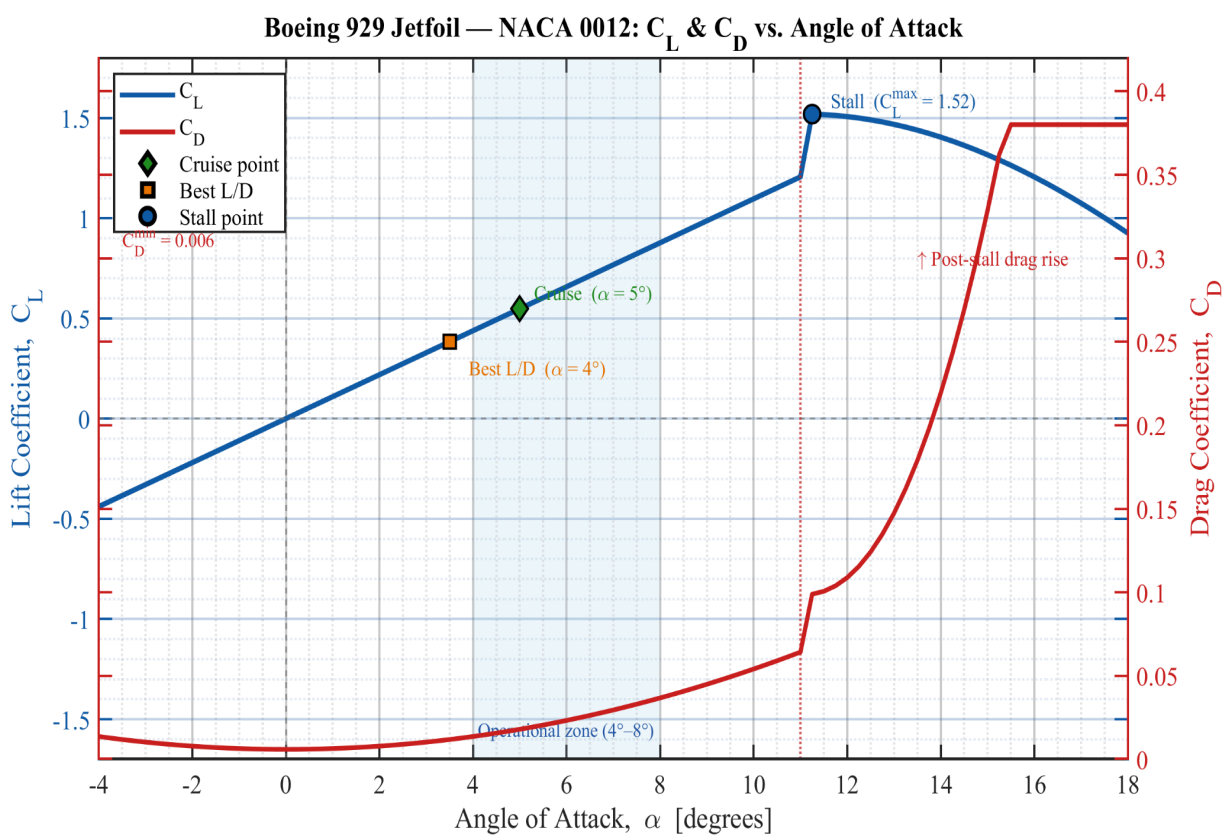
The Jetfoil's computer-controlled foils maintain optimal angle of attack, maximizing L/D across varying loads and sea states. Overall, hydrofoils excel in high-speed passenger transport where speed and comfort justify higher initial costs.



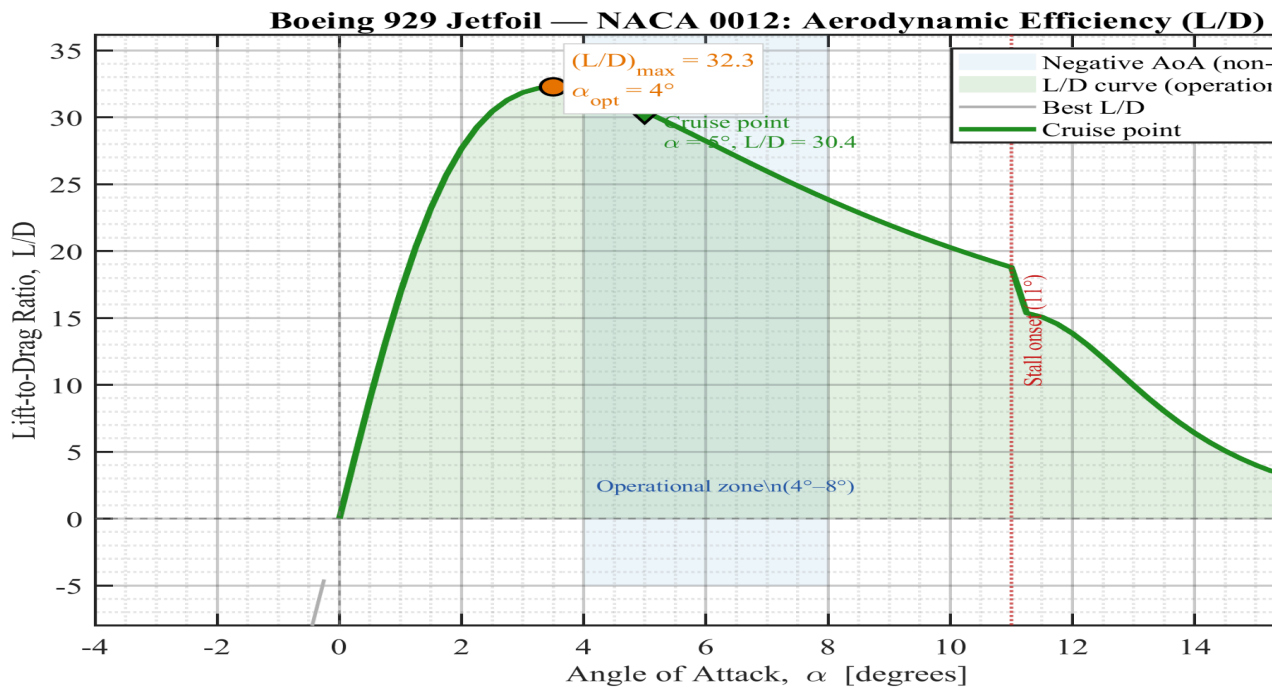
Results and Discussion (using MATLAB)

Parameter	Value
Vessel Gross Weight	961 kN (98,000 kg)
Foil Reference Area	18.0 m^2
Chord Length	1.80 m
Reynolds Number	3.72×10^7 (fully turbulent)
Minimum Drag Coefficient, C_D,min	0.0060
Maximum Lift Coefficient, C_L,max	1.519 at $\alpha = 11°$
Stall Angle	±11°
Best L/D Ratio	32.3 at $\alpha = 4°$

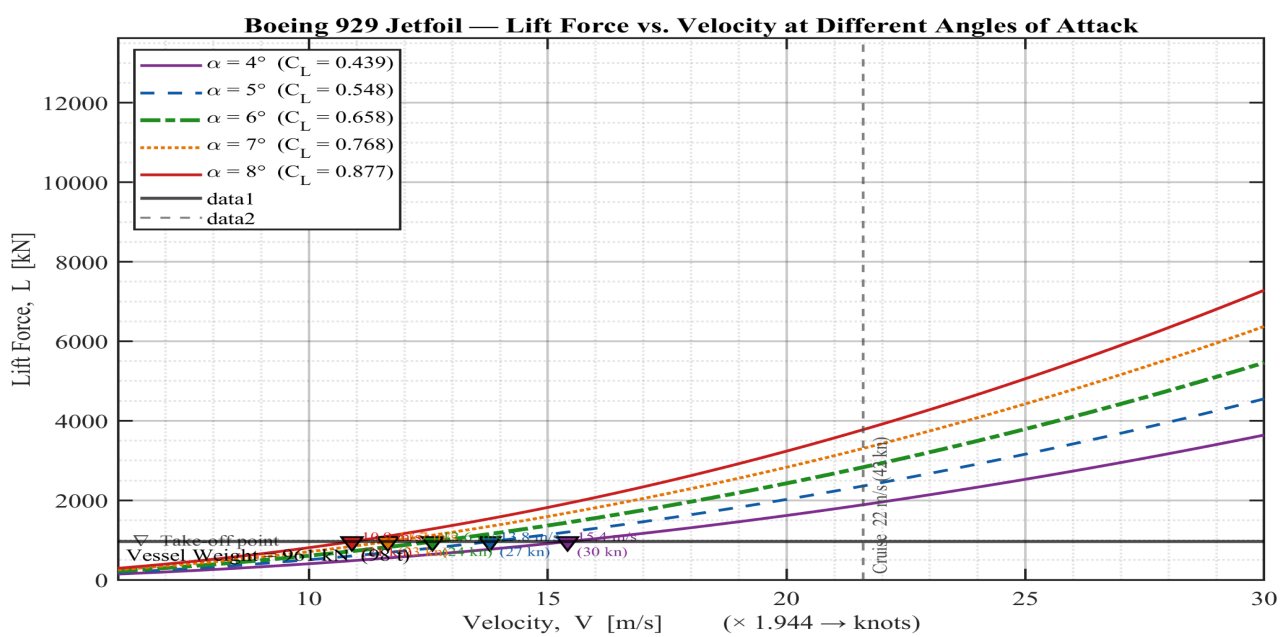
Cruise Speed	21.6 m/s (42 knots)
Cruise Angle of Attack	5.0° (design trim)
L/D at Cruise	30.4
Take-off Speed (at α = 6°)	12.59 m/s (24.5 knots)



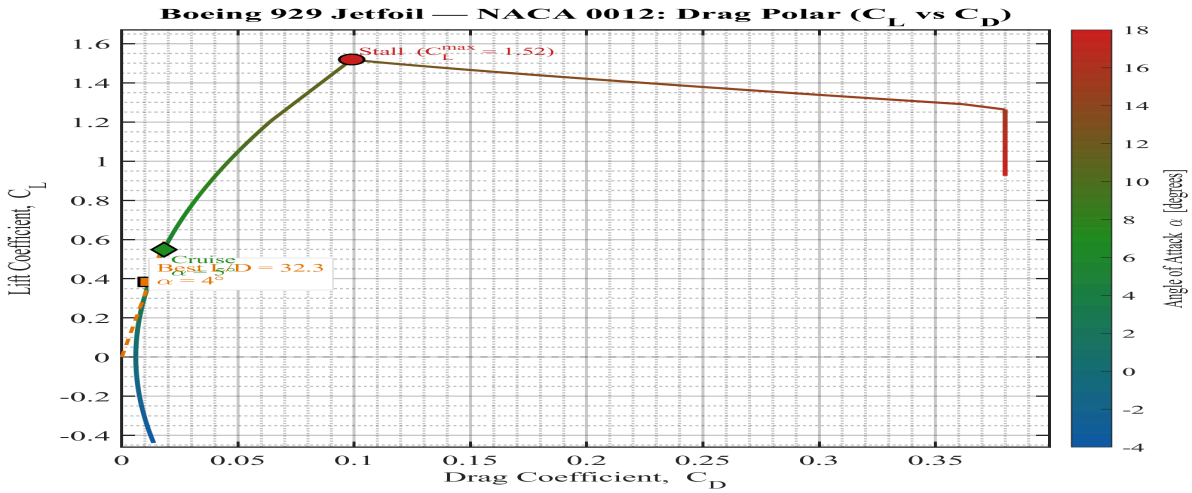
Lift and drag coefficients for NACA 0012 hydrofoil vs. angle of attack. Stall occurs at α = ±11°, with C_L,max = 1.519. Cruise condition at α = 5° lies within the optimal operating zone (4°–8°).



Aerodynamic efficiency (L/D) vs. angle of attack. Maximum L/D = 32.3 at α = 4°. Cruise point (α = 5°) gives L/D = 30.4, confirming efficient operation near peak efficiency.

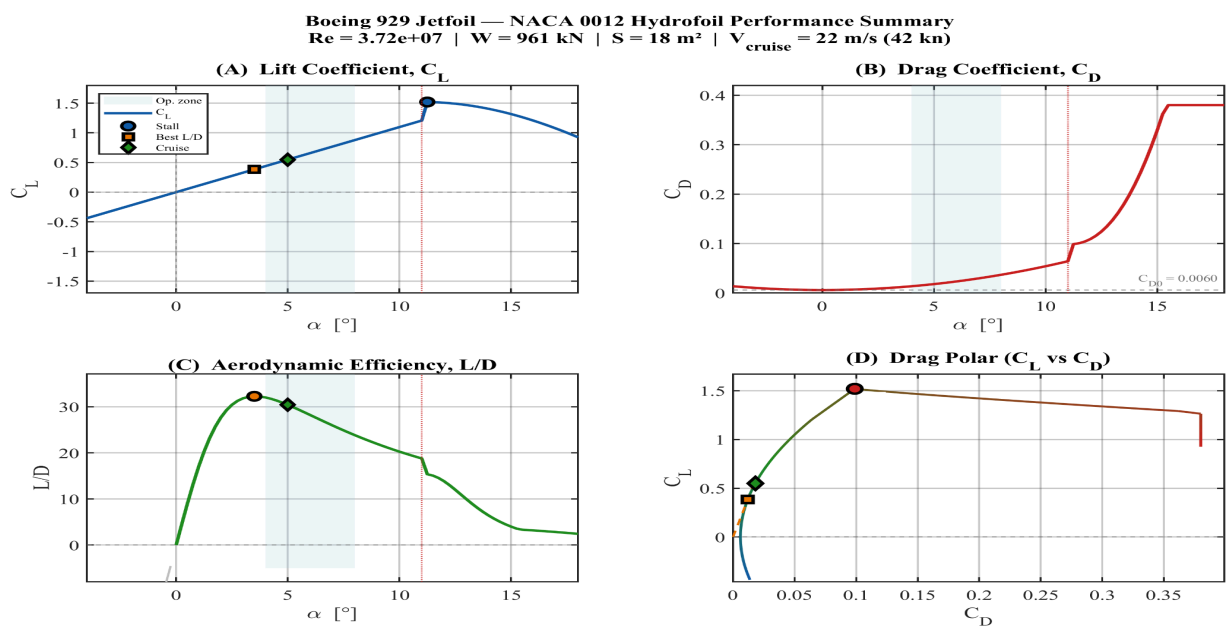


Lift force vs. vessel velocity at angles of attack 4°–8°. Take-off occurs at 12.6 m/s (24.5 knots) for α = 6°. Cruise speed is 21.6 m/s (42 knots).



Drag polar (C_L vs. C_D) for NACA 0012. Best L/D = 32.3 at α = 4°. Minimum profile drag C_D,min = 0.006, confirming low drag at high Reynolds number (Re = 3.72 × 10^7).

Performance Summary



Full performance summary - C_L, C_D, L/D, and drag polar for the Boeing 929 Jetfoil NACA 0012 hydrofoil. Key results: stall at α = 11°, best L/D = 32.3, cruise at α = 5° with L/D = 30.4.

Conclusion

Hydrofoil craft, as demonstrated by the Boeing 929 Jetfoil case study, offer superior performance through effective lift generation and drag reduction. It supports large displacements at high speeds with favourable L/D ratios. Efficiency gains make hydrofoils ideal for fast, low-emission transport, though challenges in cost and complexity remain. Future developments in materials and controls will further enhance viability.

References

1. Boeing Marine Systems technical data on Jetfoil performance.

2. Fluid dynamics principles from Abbott & von Doenhoff (NACA airfoil data).

3. Hydrofoil design literature on lift/drag calculations.

4. Abbott, I. H., & von Doenhoff, A. E. (1959). *Theory of wing sections*. Dover Publications.

5. Ladson, C. L., et al. (1988). *Compilation and analysis of high-Reynolds-number airfoil data*. NASA TM-100526.

6. Sheldahl, R. E., & Klimas, P. C. (1981). *Aerodynamic characteristics of seven symmetric airfoil sections through 180-degree angle of attack*. Sandia National Labs, SAND80-2114.

6. XFOIL documentation — Open-source airfoil analysis tool.

Code Base - Github